Solow Model — Exam Cheatsheet

14E022 Macroeconomics I, BSE

Lecture 1

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1 Notation and Definitions

Symbol	Meaning	Definition
Y_t	Aggregate output	
K_t	Aggregate capital	
L_t	Labor (population)	$L_t = L_0 e^{nt}$
A_t	Technology level	$A_t = A_0 e^{gt}$
k_t	Per-worker capital	$k_t = K_t/L_t$
y_t	Per-worker output	$y_t = Y_t/L_t = f(k_t)$
c_t	Per-worker consumption	$c_t = (1-s)\, y_t$
$\tilde{k}_t$	Per-effective-worker capital	$\tilde{k}_t = K_t/(A_t L_t)$
$\tilde{y}_t$	Per-effective-worker output	$\tilde{y}_t = Y_t/(A_t L_t) = f(\tilde{k}_t)$
s	Saving rate	$s \in (0,1)$, exogenous
δ	Capital depreciation rate	$\delta > 0$
n	Population growth rate	$\dot{L}/L = n$
g	Technology growth rate	$\dot{A}/A = g$
α	Capital share (Cobb-Douglas)	$\alpha \in (0,1)$
k^*	Steady-state per-worker capital	
k_g^*	Golden Rule capital	

Variable Transformation: Key Identity

For any variable X_t growing with labor L_t, define $x_t = X_t/L_t$. Then:

$$\frac{\dot{x}_t}{x_t} = \frac{\dot{X}_t}{X_t} - \frac{\dot{L}_t}{L_t} = \frac{\dot{X}_t}{X_t} - n$$

For per-*effective*-worker variables $\tilde{x}_t = X_t/(A_t L_t)$:

$$\frac{\dot{\tilde{x}}_t}{\tilde{x}_t} = \frac{\dot{X}_t}{X_t} - n - g$$

2 Model Setup and Assumptions

Production Function

The aggregate production function $Y_t = F(K_t, L_t)$ satisfies:

A1. Constant Returns to Scale (CRS): $F(\lambda K, \lambda L) = \lambda F(K, L)$ for all $\lambda > 0$.

A2. Positive marginal products: $F_K > 0$, $F_L > 0$.

A3. Diminishing marginal products: $F_{KK} < 0$, $F_{LL} < 0$.

A4. Inada Conditions:

$$\lim_{K\to 0} F_K = \infty, \quad \lim_{K\to\infty} F_K = 0$$
$$\lim_{L\to 0} F_L = \infty, \quad \lim_{L\to\infty} F_L = 0$$

By CRS, we can write $Y = L \cdot F(K/L, 1) \equiv L \cdot f(k)$, so:

$$y = f(k), \quad f'(k) > 0, \quad f''(k) < 0$$

with $\lim_{k\to 0} f'(k) = \infty$ and $\lim_{k\to\infty} f'(k) = 0$.

Behavioral Equations

- **Saving/Investment:** $I_t = sY_t$ (constant fraction s of output is saved)
- **Consumption:** $C_t = (1-s)Y_t$
- **Capital accumulation:** $\dot{K}_t = I_t - \delta K_t = sY_t - \delta K_t$
- **Labor:** $L_t = L_0 e^{nt}$ (exogenous growth)

3 Core Derivations

3.1 Law of Motion for Per-Worker Capital

Goal: Derive $\dot{k}_t$ from the aggregate capital accumulation equation.
Step 1. Differentiate $k_t = K_t/L_t$ with respect to time:

$$\dot{k}_t = \frac{\dot{K}_t}{L_t} - \frac{K_t \dot{L}_t}{L_t^2} = \frac{\dot{K}_t}{L_t} - nk_t$$

Step 2. Substitute $\dot{K}_t = sF(K_t, L_t) - \delta K_t$ and divide by L_t:

$$\frac{\dot{K}_t}{L_t} = s\,\frac{F(K_t, L_t)}{L_t} - \delta\,\frac{K_t}{L_t} = sf(k_t) - \delta k_t$$

Step 3. Combine:

$$\dot{k}_t = sf(k_t) - \delta k_t - nk_t$$

Law of Motion for Per-Worker Capital:

$$\boxed{\dot{k}_t = sf(k_t) - (n+\delta)\,k_t}$$

- $sf(k_t)$: gross investment per worker
- $(n+\delta)k_t$: *break-even investment* (must cover depreciation + dilution from new workers)

3.2 Steady-State Condition

At the steady state, $\dot{k}_t = 0$:

$$\boxed{sf(k^*) = (n+\delta)\,k^*}$$

The Inada conditions guarantee existence and uniqueness of $k^* > 0$.

3.3 Law of Motion in Per-Effective-Worker Terms

When technology grows at rate g, we use $\tilde{k}_t = K_t/(A_t L_t)$.

Step 1. Differentiate $\tilde{k} = K/(AL)$:

$$\dot{\tilde{k}} = \frac{\dot{K}}{AL} - \frac{K(\dot{A}L + A\dot{L})}{(AL)^2} = \frac{\dot{K}}{AL} - (g+n)\tilde{k}$$

Step 2. Since $\dot{K} = sF(K, AL) - \delta K = sAL\, f(\tilde{k}) - \delta AL\tilde{k}$:

$$\frac{\dot{K}}{AL} = sf(\tilde{k}) - \delta\tilde{k}$$

Step 3. Combine:

$$\dot{\tilde{k}} = sf(\tilde{k}) - (n+g+\delta)\,\tilde{k}$$

Law of Motion (per-effective-worker):

$$\boxed{\dot{\tilde{k}} = sf(\tilde{k}) - (n+g+\delta)\,\tilde{k}}$$

Steady state: $sf(\tilde{k}^*) = (n+g+\delta)\,\tilde{k}^*$

On the BGP, $\tilde{k}^*$ is constant, so:

$$\gamma_k = g, \quad \gamma_y = g, \quad \gamma_Y = n+g$$

4 Cobb-Douglas Special Case

Assume $f(k) = k^\alpha$ with $\alpha \in (0,1)$, so $Y = K^\alpha L^{1-\alpha}$.

Steady-State Values

From $sk^{*\alpha} = (n+\delta)k^*$:

$$sk^{*\alpha-1} = n+\delta \implies k^{*\alpha-1} = \frac{n+\delta}{s} \implies k^* = \left(\frac{s}{n+\delta}\right)^{1/(1-\alpha)}$$

Cobb-Douglas Steady State:

$$\boxed{k^* = \left(\frac{s}{n+\delta}\right)^{\frac{1}{1-\alpha}}} \qquad y^* = \left(\frac{s}{n+\delta}\right)^{\frac{\alpha}{1-\alpha}}$$

With technology:

$$\tilde{k}^* = \left(\frac{s}{n+g+\delta}\right)^{\frac{1}{1-\alpha}}$$

Comparative Statics

Parameter change	Effect on k^*	Effect on y^*
$s \uparrow$	$\uparrow$	$\uparrow$
$n \uparrow$	$\downarrow$	$\downarrow$
$\delta \uparrow$	$\downarrow$	$\downarrow$
$\alpha \uparrow$	$\uparrow$	$\uparrow$

MRW Log-Linearized Income Equation

Taking logs of $y^* = A_0 e^{gt} \cdot (\tilde{k}^*)^\alpha$ with Cobb-Douglas:

$$\ln\left(\frac{Y}{L}\right) = \ln A_0 + gt + \frac{\alpha}{1-\alpha}\ln s - \frac{\alpha}{1-\alpha}\ln(n+g+\delta)$$

5 The Golden Rule

Golden Rule Capital Level

At steady state, consumption per worker is:

$$c^* = (1-s)\,f(k^*) = f(k^*) - sf(k^*) = f(k^*) - (n+\delta)\,k^*$$

The *Golden Rule* maximizes c^* over the choice of k^* (equivalently, over s).

$$\frac{dc^*}{dk^*} = f'(k^*) - (n+\delta) = 0$$

Golden Rule Condition:

$$\boxed{f'(k_g^*) = n + \delta}$$

Equivalently, the marginal product of capital equals the break-even investment rate.
Under Cobb-Douglas: $f'(k) = \alpha k^{\alpha-1}$, so:

$$\alpha(k_g^*)^{\alpha-1} = n + \delta \implies k_g^* = \left(\frac{\alpha}{n+\delta}\right)^{1/(1-\alpha)}$$

The Golden Rule saving rate is $\boxed{s_g = \alpha}$ (equals the capital share).

Dynamic Efficiency

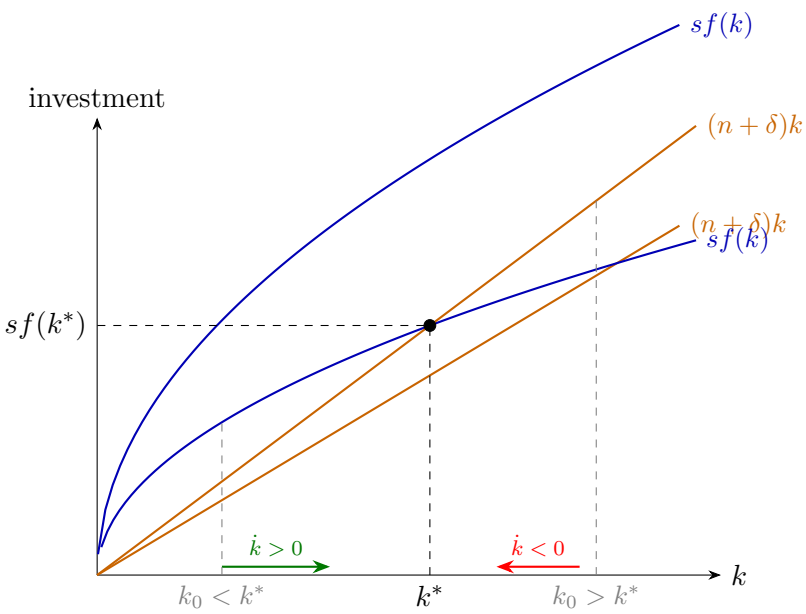
- $s < s_g = \alpha$: $k^* < k_g^*$ — **dynamically efficient** (underaccumulation). Increasing s raises c^*.
- $s > s_g = \alpha$: $k^* > k_g^*$ — **dynamically inefficient** (overaccumulation). Decreasing s raises c^* permanently.
- $s = s_g$: economy is at the Golden Rule.

Test for dynamic inefficiency: Compare s to α (Cobb-Douglas).
Equivalently: check if $\delta K >$ profits (capital income), i.e., $r < n + g$ in AK/Ramsey context.

6 Transitional Dynamics and Phase Diagram

Phase Diagram

The phase diagram plots $sf(k)$ (gross investment) against $(n+\delta)k$ (break-even investment).



Reading the Phase Diagram

- **Above the intersection** ($k < k^*$): $sf(k) > (n+\delta)k$, so $\dot{k} > 0$ (capital accumulates).
- **Below the intersection** ($k > k^*$): $sf(k) < (n+\delta)k$, so $\dot{k} < 0$ (capital decumulates).
- The Inada conditions guarantee $sf(k) > (n+\delta)k$ for small k and $sf(k) < (n+\delta)k$ for large k.
- The steady state k^* is **globally stable**.

7 Speed of Convergence

Log-Linearization Around Steady State

Let $\hat{k}_t = \ln k_t - \ln k^*$ (log deviation from steady state).
Step 1. The law of motion is $\dot{k} = sf(k) - (n+\delta)k$.
Step 2. Define $\phi(k) \equiv \dot{k}$. At k^*, $\phi(k^*) = 0$. Taylor-expand:

$$\dot{k}_t \approx \phi'(k^*)(k_t - k^*)$$

Step 3. Compute $\phi'(k^*)$:

$$\phi'(k) = sf'(k) - (n+\delta)$$

At steady state, $sf(k^*) = (n+\delta)k^*$, so $s = (n+\delta)k^*/f(k^*)$. Thus:

$$\phi'(k^*) = (n+\delta)\frac{k^* f'(k^*)}{f(k^*)} - (n+\delta) = (n+\delta)\big[\varepsilon_{f,k}(k^*) - 1\big]$$

where $\varepsilon_{f,k}(k^*) = k^* f'(k^*)/f(k^*)$ is the elasticity of output w.r.t. capital at k^*.
Step 4. Under Cobb-Douglas, $f(k) = k^\alpha$ implies $\varepsilon_{f,k} = \alpha$ everywhere:

$$\phi'(k^*) = (n+\delta)(\alpha - 1) = -(1-\alpha)(n+\delta)$$

Speed of convergence:

$$\boxed{\beta = (1-\alpha)(n+\delta)}$$

Convergence equation:

$$\dot{\hat{k}}_t \approx -\beta \hat{k}_t \implies \hat{k}_t = \hat{k}_0\, e^{-\beta t}$$

$$\ln k_t = (1 - e^{-\beta t}) \ln k^* + e^{-\beta t} \ln k_0$$

Half-life:

$$t_{1/2} = \frac{\ln 2}{\beta}$$

Properties of Beta

- β increases with n and δ (higher dilution $\Rightarrow$ faster convergence).
- β decreases with α (higher capital share $\Rightarrow$ slower convergence; diminishing returns matter more when α is small).

- The saving rate s **does not** affect β under Cobb-Douglas (it affects k^* but not speed of convergence).
- With $\alpha \approx 1/3$, $n = 0.01$, $\delta = 0.05$: $\beta \approx 0.04$ (half-life ≈ 17 years).

Growth rate approximation:

$$\gamma_{y,t} \equiv \frac{\dot{y}_t}{y_t} \approx \alpha \cdot \frac{\dot{k}_t}{k_t} \approx -\alpha\beta(\ln k_t - \ln k^*) \approx -\beta(\ln y_t - \ln y^*)$$

Countries further below their steady state grow faster (*conditional convergence*).

8 Empirics: MRW (1992) and Extensions

Mankiw-Romer-Weil (1992) Baseline

Cross-country regression implied by Solow model:

$$\ln\left(\frac{Y_i}{L_i}\right) = a + \frac{\alpha}{1-\alpha}\ln s_i - \frac{\alpha}{1-\alpha}\ln(n_i + g + \delta) + \varepsilon_i$$

- **Basic result:** $R^2 \approx 0.6$, but implied $\alpha \approx 0.6$–0.7 (too high; true capital share $\approx 1/3$).
- **Problem:** $\alpha \approx 1/3$ requires the coefficient ratio ≈ 0.5, but data gives ≈ 1.4.

Augmented Solow: Human Capital

Augment with human capital H: $Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}$.

Let s_k, s_h denote investment rates in physical and human capital. Steady state:

$$\ln\left(\frac{Y}{L}\right) = \ln A_0 + gt + \frac{\alpha}{1-\alpha-\beta}\ln s_k - \frac{\alpha+\beta}{1-\alpha-\beta}\ln(n+g+\delta) + \frac{\beta}{1-\alpha-\beta}\ln s_h$$

MRW augmented results: $R^2 \approx 0.8$, implied $\hat{\alpha} \approx \hat{\beta} \approx 1/3$. Consistent with factor shares.

Development Accounting

Decompose cross-country income differences:

$$\ln y_i - \ln y_j = \underbrace{\alpha(\ln k_i - \ln k_j)}_{\text{capital}} + \underbrace{(1-\alpha)(\ln h_i - \ln h_j)}_{\text{human capital}} + \underbrace{(\ln A_i - \ln A_j)}_{\text{TFP}}$$

Finding: TFP differences account for ~50% of cross-country income variance.

The Lucas Puzzle

If $Y = K^\alpha L^{1-\alpha}$, the marginal product of capital is:

$$MPK = \alpha\frac{Y}{K} = \alpha(y/k)$$

Puzzle: If $y_{\text{US}}/y_{\text{India}} \approx 15$ and $\alpha = 1/3$, then $MPK_{\text{India}}/MPK_{\text{US}} \approx 15^{1-\alpha} \approx 58$. Capital should flow massively to poor countries. It doesn't.

Resolutions: Human capital differences, TFP/institutions, political risk, financial frictions.

9 Extensions: Poverty Traps and Uzawa's Theorem

Poverty Traps

Standard Solow implies a unique k^*. Non-convexities can generate multiple steady states.

Two-technology setup:

- Technology A (subsistence): $y = Ak^\alpha$ (low productivity)
- Technology B (modern): $y = Bk^\alpha - \gamma$ where $B > A$, $\gamma > 0$ (fixed cost)
- Threshold capital:

$$\tilde{k} = \left(\frac{\gamma}{B - A}\right)^{1/\alpha}$$

Poverty trap:

- $k_0 < \tilde{k}$: economy uses technology A, converges to low k_A^*.
- $k_0 > \tilde{k}$: economy uses technology B, converges to high k_B^*.
- **"Big Push":** coordinated investment to push $k_0 > \tilde{k}$.

The $sf(k)$ curve is non-concave and crosses the break-even line multiple times.

Uzawa's Theorem

Question: What restrictions does a BGP impose on the production function?

Uzawa's Theorem: On a BGP with $\gamma_Y = n + g > 0$ and $\gamma_K > 0$, the production function must be asymptotically **labor-augmenting (Harrod-neutral)**:

$$Y = F(K, AL)$$

Capital-augmenting or Hicks-neutral technical progress is inconsistent with a BGP (unless $\sigma = 1$, i.e., Cobb-Douglas).

Proof sketch: On BGP, K/Y is constant. With constant $r = F_K$, this requires F_K to depend only on K/Y or equivalently $K/(AL)$, which forces labor augmentation.

Implication: The standard form $Y = F(K, A_t L_t)$ is not merely a convention but a requirement for balanced growth.

10 Problem-Solving Templates

Template 1: Find Steady State

1. Write law of motion: $\dot{k} = sf(k) - (n + \delta)k$
2. Set $\dot{k} = 0$: $sf(k^*) = (n + \delta)k^*$
3. For Cobb-Douglas: $sk^{*\alpha} = (n + \delta)k^*$, solve for k^*
4. Find $y^* = f(k^*)$, $c^* = (1 - s)y^*$

Template 2: Comparative Statics

Use the steady-state condition implicitly:

$$sf(k^*) = (n+\delta)k^*$$

Differentiate both sides with respect to the parameter of interest. E.g., for $\partial k^*/\partial s$:

$$f(k^*) + sf'(k^*)\frac{\partial k^*}{\partial s} = (n+\delta)\frac{\partial k^*}{\partial s} \implies \frac{\partial k^*}{\partial s} = \frac{f(k^*)}{(n+\delta) - sf'(k^*)} > 0$$

(positive because the denominator equals $(n+\delta)(1-\varepsilon_{f,k}) > 0$).

Template 3: Growth Accounting

$$\frac{\dot{Y}}{Y} = \alpha\frac{\dot{K}}{K} + (1-\alpha)\frac{\dot{L}}{L} + \frac{\dot{A}}{A}$$

$$\text{TFP growth} = \frac{\dot{A}}{A} = \frac{\dot{Y}}{Y} - \alpha\frac{\dot{K}}{K} - (1-\alpha)\frac{\dot{L}}{L}$$

Template 4: Speed of Convergence Calculation

Given α, n, δ:

1. Compute $\beta = (1-\alpha)(n+\delta+g)$ (with technology growth g)
2. Half-life: $t_{1/2} = \ln(2)/\beta \approx 0.693/\beta$
3. Convergence fraction after T years: $1 - e^{-\beta T}$

Template 5: Golden Rule

1. Write $c^* = f(k^*) - (n+\delta)k^*$
2. Maximize: $f'(k^*) = n+\delta$
3. Under Cobb-Douglas: $\alpha k^{*\alpha-1} = n+\delta$, and $s_g = \alpha$
4. Check: if current $s > \alpha$, dynamically inefficient

11 Common Pitfalls

P1. Confusing $k = K/L$ vs. $\tilde{k} = K/(AL)$: The break-even rate for k is $(n+\delta)$; for $\tilde{k}$ it is $(n+g+\delta)$. Never mix them.

P2. Exponent in Cobb-Douglas steady state:

$$k^* = \left(\frac{s}{n+\delta}\right)^{1/(1-\alpha)} \quad \text{NOT} \quad \left(\frac{s}{n+\delta}\right)^{1/\alpha}$$

P3. Speed of convergence formula: $\beta = (1-\alpha)(n+\delta)$ for k, but $\beta = (1-\alpha)(n+g+\delta)$ for $\tilde{k}$.

P4. Saving rate and convergence: s does NOT affect β (Cobb-Douglas). A higher s raises k^* but not the speed of convergence.

P5. Dynamic inefficiency direction: Overaccumulation ($s > s_g = \alpha$) is dynamically *in*efficient. Underaccumulation ($s < \alpha$) is dynamically *efficient*.

P6. Golden Rule vs. Modified Golden Rule: The Solow Golden Rule is $f'(k^*) = n + \delta$. The Ramsey Modified Golden Rule is $f'(k^*) = \rho + \delta$ (different condition — do not confuse).

P7. Per-worker vs. per-capita: If some fraction of population does not work, adjust accordingly. Usually L = workers = population in Solow.

P8. BGP requires technology growth: On BGP, y grows at rate g, not at zero. Stagnant BGP only if $g = 0$.

P9. Log-linearization sign: $\dot{\hat{k}} = -\beta\hat{k}$ means $\hat{k}$ decays toward zero, confirming stability. If you get $+\beta$, check signs.

P10. Inada conditions and phase diagram: $\lim_{k\to 0} sf'(k) = \infty > n + \delta$ ensures $sf(k)$ is above $(n + \delta)k$ near origin. Always verify this for non-Cobb-Douglas cases.

12 Quick Reference Formulas

Laws of Motion:

$$\dot{k} = sf(k) - (n+\delta)k \qquad \text{(per worker)}$$
$$\dot{\tilde{k}} = sf(\tilde{k}) - (n+g+\delta)\tilde{k} \qquad \text{(per eff. worker)}$$

Steady States (Cobb-Douglas $f(k) = k^\alpha$):

$$k^* = \left(\tfrac{s}{n+\delta}\right)^{1/(1-\alpha)}, \quad y^* = \left(\tfrac{s}{n+\delta}\right)^{\alpha/(1-\alpha)}$$
$$\tilde{k}^* = \left(\tfrac{s}{n+g+\delta}\right)^{1/(1-\alpha)}$$

Golden Rule:

$$f'(k_g^*) = n+\delta, \quad s_g = \alpha \text{ (Cobb-Douglas)}$$

Speed of Convergence:

$$\beta = (1-\alpha)(n+\delta), \quad t_{1/2} = \frac{\ln 2}{\beta}$$

MRW Income Equation:

$$\ln(Y/L) = \ln A_0 + gt + \frac{\alpha}{1-\alpha}\ln s - \frac{\alpha}{1-\alpha}\ln(n+g+\delta)$$

MRW Augmented (with human capital β_H):

$$\ln(Y/L) \propto \frac{\alpha}{1-\alpha-\beta_H}\ln s_k + \frac{\beta_H}{1-\alpha-\beta_H}\ln s_h - \frac{\alpha+\beta_H}{1-\alpha-\beta_H}\ln(n+g+\delta)$$

Growth Accounting:

$$\dot{A}/A = \dot{Y}/Y - \alpha\,\dot{K}/K - (1-\alpha)\,\dot{L}/L$$

Poverty Trap Threshold:

$$\tilde{k} = \left(\frac{\gamma}{B-A}\right)^{1/\alpha}$$

Variable	Per worker	Aggregate
K/L, Y/L, C/L	g	$n+g$
r (return to capital)	0	0
w (real wage)	g	$n+g$